

AI Chatbot – Task 2: NLP-Based Chatbot using TF-IDF

Introduction

This project focuses on developing an NLP-based chatbot capable of answering college-related queries using Natural Language Processing techniques. Unlike a rule-based chatbot, this system understands user queries based on text similarity using the TF-IDF Vectorizer and Cosine Similarity. The chatbot preprocesses user input, converts it into numerical vectors, compares it with stored questions, and returns the most relevant response. This approach improves flexibility and enables the chatbot to understand different ways of asking the same question.

Objective

The objective of this project is to build an intelligent chatbot that can understand natural language queries using TF-IDF and Cosine Similarity. The chatbot should accurately identify user intent, provide relevant responses, and improve interaction compared to simple keyword matching. It should also handle different sentence structures while maintaining fast and accurate responses.

Libraries Used

- Python
- NLTK
- Scikit-learn
- NumPy
- Pickle
- TF-IDF Vectorizer
- Cosine Similarity

Methodology

The chatbot first preprocesses the dataset by converting text to lowercase, removing punctuation, and tokenizing sentences. A TF-IDF Vectorizer transforms the text into numerical feature vectors. When the user enters a query, the same preprocessing is applied, and Cosine Similarity is calculated between the user's input and stored questions. The chatbot selects the response with the highest similarity score. The trained TF-IDF model is saved as a .pkl file for future use, and a confusion matrix is generated to evaluate chatbot performance.

Result

The NLP chatbot successfully answered user questions related to admissions, fees, courses, hostels, placements, greetings, and other college information. The chatbot correctly identified user intent using text similarity and generated relevant responses. The trained TF-IDF model and confusion matrix demonstrated the effectiveness of the chatbot.

Conclusion

The NLP-based chatbot was successfully developed using Python, NLTK, TF-IDF Vectorizer, and Cosine Similarity. Compared to the rule-based chatbot, it provides more flexible and intelligent responses by understanding the meaning of user queries instead of relying only on predefined keywords. This project demonstrates the practical application of Natural Language Processing for building smart conversational AI systems.